

Hostel Outing Permission & Approval Management System

User Guide

Welcome to the end-user manual for the **Hostel Outing Permission & Approval Management System**. This guide provides step-by-step instructions for Students, Heads of Department (HOD), Hostel Wardens, and Main Gate Watchmen.

1. Student Manual

1.1 Account Login & Registration

1. **Access Portal:** Open `http://localhost:5173` in your web browser.
2. **Login:** Enter your student email address (e.g., `student.a@hostelapp.local`) and password.
3. **Sign Up (New Students):**
 - o Click **Sign Up** on the login screen.
 - o Select role **Student**.
 - o Fill in Required Fields: Full Name, Email, Password, Confirm Password, Register Number (e.g. CSE2027001), Department (e.g. Computer Science and Engineering), Hostel Block (e.g. A Block), Academic Year, and Room Number.
 - o Click **Create Account**.

1.2 Student Dashboard Overview

Upon logging in, your dashboard displays:

- **Student Profile Summary:** Name, Register Number, Department, Hostel Block, Room Number.
- **Active Outings Section:** Cards showing your current active outing requests.
- **Action Buttons:** + New Outing Request and View Personal History.

1.3 Creating an Outing Request

1. Click + **New Outing Request**.
2. Complete the required fields:

- **Outing Date:** Select scheduled date (cannot be in the past).
 - **Leaving Time:** Scheduled departure time.
 - **Expected Return Time:** Scheduled return time (must be after leaving time).
 - **Destination:** Destination name/address.
 - **Reason:** Specific purpose of outing.
3. Click **Submit Outing Request**.
 4. The request status is initialized to `PENDING_HOD`.

1.4 Viewing Request Status & Tracking Progress

Your request progresses through visual status badges:

- **PENDING_HOD:** Submitted; awaiting academic approval by your HOD.
- **PENDING_WARDEN:** HOD approved; awaiting parent confirmation and final approval by your Hostel Warden.
- **APPROVED:** Warden granted final approval; ready for gate exit.
- **EXITED:** You have exited the campus gate.
- **COMPLETED:** Returned to hostel on or before expected return time.
- **LATE_RETURN:** Returned to hostel after expected return time.
- **REJECTED:** Declined by HOD or Warden.
- **CANCELLED:** Cancelled by you.

1.5 Parent Approval Status

In `PENDING_WARDEN` status, your Warden will contact your parent/guardian to verify permission. You can track whether parent confirmation has been completed by checking the **Parent Approval Confirmed** flag on your request card or history timeline.

1.6 Cancelling Eligible Requests

You can cancel an outing request as long as its status is `PENDING_HOD` or `PENDING_WARDEN`:

1. Locate the request card on your dashboard.
2. Click **Cancel Request**.
3. Status updates to `CANCELLED`, releasing any active time window blocks.

1.7 Viewing Personal Outing History

1. Click **View History** on your dashboard.
2. Browse past outing requests and click **View Audit Timeline** to inspect the detailed history log (showing exact submission, approval, exit, and return timestamps).

2. HOD (Head of Department) Manual

2.1 Login & Department Scope Rule

- Log in with your HOD credentials (e.g., `hod.cse@hostelapp.local`).
- **Department Scope Rule:** You can **ONLY** view and process outing requests for students enrolled in your assigned department. Access to other departments is strictly forbidden by the system.

2.2 Department Pending Dashboard

1. The HOD Dashboard (`/hod`) displays all pending requests (`PENDING_HOD`) for your department.
2. Review student information: Name, Register Number, Year, Destination, Outing Date, Times, and Reason.

2.3 Approving & Rejecting Requests

- **To Approve:**
 1. Optionally enter an academic review note in the comment box.
 2. Click **Approve**.
 3. Status advances to `PENDING_WARDEN`, routing the request to the student's Hostel Warden.
- **To Reject:**
 1. Enter a rejection reason in the comment box.
 2. Click **Reject**.
 3. Status changes to `REJECTED`, terminating the workflow.

2.4 Department History & Audit Timeline

1. Navigate to **HOD History** (`/hod/history`).
 2. View departmental statistics (Total Outings, Approved, Rejected, Late Returns).
 3. Search by student name, register number, or outing ID `#OUT-X`.
 4. Filter by status (`APPROVED`, `COMPLETED`, `LATE_RETURN`, etc.) or hostel block.
 5. Click **View Audit Timeline** on any record to inspect exact timestamps and actor comments.
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3. Warden Manual

3.1 Login & Hostel Block Scope Rule

- Log in with your Warden credentials (e.g., `warden.a@hostelapp.local`).
- **Hostel Block Scope Rule:** You can **ONLY** view and process outing requests for students residing in your assigned hostel block. Access to other hostel blocks is strictly forbidden by the system.

3.2 Warden Pending Dashboard

1. The Warden Dashboard (`/warden`) displays HOD-approved requests (`PENDING_WARDEN`) for your hostel block residents.

2. Review student details, room number, destination, and expected return time.

3.3 Mandatory Parent Approval Confirmation

The system enforces parent/guardian consent verification:

1. Contact the student's parent or legal guardian via phone or in-person.
2. Check the mandatory box: `[x] Parent / Guardian Approval Confirmed by Phone`.
3. Click **Confirm Parent Approval**.
4. The system logs a `PARENT_APPROVAL_CONFIRMED` audit event.

3.4 Granting Final Approval or Rejection

- **To Approve:**
 1. Ensure parent confirmation has been completed.
 2. Optionally enter a Warden comment.
 3. Click **Approve Outing**. Status changes to `APPROVED`.
- **To Reject:**
 1. Enter a reason for rejection (e.g., "Parent declined consent").
 2. Click **Reject Outing**. Status changes to `REJECTED`.

3.5 Hostel Block History & Late Return Tracking

1. Navigate to **Warden History** (`/warden/history`).
 2. Review block-wide metrics including Late Return counts.
 3. Filter by department or search for specific student records.
 4. Click **View Audit Timeline** to inspect gate movement and return delay details.
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4. Watchman Manual

4.1 Login & Main Gate Desk

- Log in with Watchman credentials (`watchman@hostelapp.local`).
- The Watchman Gate Desk (`/watchman`) provides campus-wide gate verification operations.

4.2 Searching Students & Verifying Approval

1. Use the **Search Bar** to lookup students by:
 - Student Register Number (e.g. `CSE2027001`).
 - Student Name (e.g. `Arjun`).
 - Outing Request ID (e.g. `#OUT-4`).
2. Verify that the student's current status is strictly **APPROVED** before allowing campus exit.

4.3 Recording Gate Exit

1. Locate the student's outing card under **Today's Active Outings**.
2. Confirm the student's identity and destination.
3. Click **Record Exit**.
4. The status updates to **EXITED**, and the gate exit timestamp is recorded.

4.4 Recording Gate Return

1. When the student arrives back at the gate, locate their record (**EXITED** status).
2. Click **Record Return**.
3. The system automatically records the return timestamp and evaluates return punctuality.

4.5 Understanding Completed vs. Late Return Statuses

- **COMPLETED**: The student returned on or before their expected return time.
 - **LATE_RETURN**: The student returned after their expected return time. The system logs the delay minutes and displays a warning toast.
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5. Late Return Detection Manual

The system automates late return compliance evaluation:

5.1 Business Logic & Formula

When the Watchman clicks **Record Return**:

- `expected_return_datetime = outing_date + expected_return_time`
- `actual_return_datetime = current_timestamp`

$$\text{is_late} = \text{actual_return_datetime} > \text{expected_return_datetime}$$
$$\text{delay_minutes} = \max\left(1, \left\lfloor \frac{\text{actual_return_datetime} - \text{expected_return_datetime}}{60} \right\rfloor\right)$$

5.2 Example Scenarios

- **Scenario A (On Time):**
 - Expected Return: 5:00 PM
 - Actual Return: 4:48 PM
 - Result: Status = **COMPLETED**, Delay = 0 minutes.
- **Scenario B (Late Return):**

- Expected Return: 5:00 PM
- Actual Return: 5:35 PM
- Result: Status = **LATE_RETURN**, Delay = 35 minutes.
- Audit Trail: Appends LATE_RETURN_DETECTED action with "Expected Return: 05:00 PM | Actual Return: 05:35 PM | Delay: 35 minutes".

*[!IMPORTANT] **Campus Entry Rule:** A student returning late is **NEVER blocked** from entering campus. The Watchman records the return, campus entry is granted immediately, and the system logs the late return for HOD, Warden, and Student administrative visibility.*